The Periodic Table of Waves v2 — Particle Classification by Winding Addresses and Observation Clocks, and the Unification of Mass, Lifetime, and Splitting by the Clock Field ω(x)

Author: Noriaki Kihara **Date**: August 7, 2026 **Type**: Fully revised hypothesis-proposal paper (supported by numerical experiments; not a proof paper) **Relation between versions**: This paper is a full revision of v1 [21] and is self-contained — it can be read without v1. All claims of v1 (the five pillars) are carried over without weakening, and the new findings obtained by the post-publication verification program (new pillars 6–10 and the consistency battery) are added as Part II.

Abstract

What classifies a particle? From numerical experiments on wave dynamics that assume no concept of a particle (two already-published systems, used as they are), this paper proposes the **"periodic table of waves"** — a classification hypothesis based on winding-number addresses and observation clocks. The starting point was a sign-vector thought experiment [1] that classified particles by which of the six axes (x, y, z, t, R, Q) carries their winding. This paper rebuilds that classification purely from the shapes of waves, by measurement, and arrives at five pillars. **Pillar 1 (periodic law)**: the foundation of the classification is a measured universal clock — the collective clock $\omega = \pi/72$/step holds across the entire range $N = 4$ to 144 (long window ±0.1%) — and species are classified by rational addresses on a single U^144 clock lattice and their relation to the clock. **Pillar 2 (charge law)**: charge is Q_em = m/3 (m the raw winding number); the dominant observation clock (denominator 3 — based on the previously measured tongue-width ratio of 461) reads "3 windings = one elementary charge". Consequently the u type m=+2 reads +2/3, the d type m=−1 reads −1/3, the electron m=−3 reads −1, and **confinement and fractional charge become two faces of the single fact of "mod-3 readability"** (the arithmetic checks pass automatically: proton uud = +3 → +1, neutron udd = 0). In the numerical experiments, quark-type addresses remain strictly unreadable in isolation (readable fraction 0.0000 persists), while in the sea they become readable, in integer charge units only, exactly to the extent that hadronization (formation of m≡0 composites) proceeds. **Pillar 3 (statistics law)**: the fermion/boson distinction lives not in spatial rotation but in the **double cover of the clock**. The state phase is quantized to {0, π} at each observable recurrence (measured to 3 digits); fermionic species (odd harmonics) visit both sheets (covering degree 2) and bosonic species do not (covering degree 1). A pure m=0 species in the fermionic band also shows covering degree 2 — the row of the neutral fermion (neutrino) is established. **Pillar 4 (mass law)**: mass is not an attribute of the address but a relational quantity with the sea (vacuum) — isolated addresses are all unitarily equivalent by winding-shift symmetry (measured), and in-sea properties depend only on the divisor class gcd(m, 16) (odd addresses match to 5 digits). **Pillar 5 (the periodic table)**: the table is a pair — the isolated stable-element table and the in-sea lifetime table — on which an assignment of the 62 Standard-Model species is proposed.

v2 goes one step further (Part II, new pillars 6–10). **New pillar 6 (the one-line law)**: the dynamical core of the classification is bound into a single formula, **$r = \sin^2\theta = P_{odd}/(P_{odd} + P_{even})$** — the reflection rate per interaction equals the fraction of odd-harmonic (fermionic-band) power. The pure bosonic sea is an exact fixed point with $r = 0$ (the vacuum does not tick); the source that drives the clock is fermionic-band content only; and the necessary condition for the Z_2 classification to have dynamical meaning in this model is precisely this asymmetric coupling (equal-amplitude $f = 0.500000$ is not a fixed point but flows to $f^* = 0.4936 \pm 0.0012$ — measured). **New pillar 7 (the spatial double-cover theorem)**: the even/odd distinction is the Z_2 representation of the half-translation T: $x \to x + n/2$ of the fundamental wavelength; pure-parity species cannot localize at a single point and necessarily carry an antipodal two-point structure — the double cover

of the clock is realized as spatial structure. A point-localized body necessarily mixes both parities and carries no statistics. **New pillar 8 (unification of mass, lifetime, and generations)**: for the distribution of the local clock field ω(x) on the support of a species, mass = ⟨ω⟩ (the mean) and lifetime = $1/\sigma_\omega$ (the inverse width); the linewidth-lifetime relation $\tau_{coh}\cdot\sigma_\omega \approx 32$ (CV 29.7%, over a 6× range of σ_ω, measured) holds. Mass and linewidth are positively correlated (+0.904, measured) — **"the heavier, the shorter-lived" follows automatically as two moments of one distribution**. From this we propose a new candidate for generations: "generations = a discrete series of phase-lock levels". **New pillar 9 (the splitting readout)**: the splitting (decay) of a wave requires no added force — dephasing by non-uniform ω *is* the dynamics of splitting; all that is needed is a readout criterion (accumulated relative phase exceeding π = two particles; $t_{split} = \pi/\Delta\omega$; measured $t\cdot\Delta\omega/\pi = 1.29$). **New pillar 10 (the two conditions of a stationary particle)**: a true stationary particle of the dynamics requires both localization (stacking all harmonics of the fundamental wavelength, center-phased — ringing decreases monotonically toward zero, a 50× improvement measured) and phase locking (ω(x) = const, a nonlinear eigenvalue problem = the candidate particle equation). Together, the classification of species, particle number, mass, lifetime, and splitting, and the gauges of space and time, are unified as **outputs of a single readout function** (the readout unification hypothesis).

v2 additionally reports a consistency battery (Part II, §14): nine pillar experiments were run, with scripts unmodified, against two localized readout engines in parallel, confirming that the integer and phase structures (Z_2 covering, cross table, neutrino row, exact confinement zero, ledger identity, address equivalence, divisor classes) are completely robust — the classification is not an artifact of a particular engine. Under this hypothesis, the problem left open by the previous paper [5] — "charge does not stabilize at ±1" — turns from a contradiction into a prediction, the 1/3-charge problem of quarks connects to confinement through a single mechanism, and the Standard-Model pattern "heavier generations are shorter-lived" acquires a structural candidate explanation. This paper claims no proof. What it claims is that twelve independently measured structures connect, under a single classification hypothesis, the particle structure of the Standard Model, the open problem of the previous paper, and the sign-vector thought experiment, all at once. All experiments are published as deterministic scripts, and the entire process, including fifteen refutations, is reproducible. The closing section extends the requirements that the universal interaction function of the next paper must satisfy, from 8 items to 11.

Part I — The Classification Hypothesis (the five pillars of v1, carried over in full)

1. Introduction — why a "periodic table", and why a hypothesis proposal

1.1 Starting point: the sign-vector thought experiment

A preceding thought experiment [1] proposed classifying particles by which of the six axes (x, y, z, t, R, Q) carries their winding — photon γ=(0,0,0,0,R,0), W±=(0,0,0,±t,0,0), gluon g=(0,0,0,0,0,Q), graviton G=(0,0,0,±t,±R,0). The classification is attractive but remained at the level of a thought experiment. Can particle species be classified as **measurements** of wave dynamics — that is the question of this paper.

1.2 The central proposition

All pillars of this paper are bound by one proposition:

> **The species of a particle is not a fixed attribute of the wave itself; it is classified by how an observation clock reads the winding state.** Charge, confinement, statistics, mass, and lifetime are not independent labels — they decompose into **the state-side address × the observation clock × the relation to the sea**.

Charge quantization exists neither in the state space alone nor in the measuring device alone — it exists in **the relation between the cyclic structure of the state and the divisor structure of the clock** (§4). Statistics lives in the double cover of the clock, not in spatial rotation (§5), and mass is differentiated by the sea (§6). v2 deepens the proposition by one step:

> **(v2 extension) The readout that classifies species and the readout that creates the gauges of space and time are one and the same relational projection.** Particle number, mass, lifetime, and splitting are unified as outputs of that readout function (Part II).

1.3 Declaration of method: no particles assumed

This paper introduces no new dynamics. It uses the two already-published systems — the N-body relational-wave dynamics [2] (used for the measurements on three-direction condensates) and the closed-form exact solution of the two-body universal inelastic map [3] (used for all measurements of charge, statistics, and confinement) — both built on the anonymity principle: "only the shape of the wave is input, and there is no special-casing inside." The concepts of particle, charge, and spin are not in the dynamics. We measure how they arise **as readouts**. (Part of the measurements in Part II are performed on a localization prototype of the readout — §2.4; that the prototype preserves the classification of this paper is confirmed by the consistency battery of §14.)

1.4 Declaration of character, and historical lessons

This is a **hypothesis-proposal paper**. The enterprise of classifying particles has a history of both success and failure. Gell-Mann's Eightfold Way [9] predicted the Ω^- from SU(3) classification and succeeded. Kelvin's vortex atoms [8] — atoms as knotted vortices, a topological classification — were beautiful, but the dynamics was wrong. Preon models [14] attempted full assignments at the 62-species scale and declined. **A beautiful classification does not guarantee correct dynamics.** That this paper remains a hypothesis proposal, strictly separates readout-level claims from dynamics-level claims (§2.3), and states its refutation conditions (§17), is due to this lesson. The value of the hypothesis should be measured by the breadth of its connections and the clarity of its refutation conditions.

2. Methods and conventions

2.1 Dynamics (read-only)

- **N-body system** [2]: closure dynamics of N relational waves (edges of the complete graph). Forms three-direction metastable condensates. Used for the clock universality scan (§3).
- **Two-body system** [3]: closed-form exact solution of the two-channel wave (a, b) with point-wise interaction $\delta a = -2R\cdot\mathrm{Im}(\bar{b}a)\cdot b$. States live on a (χ, η) lattice; the η winding number m is the object of the charge readout. Used for all experiments on charge, statistics, and confinement (§4–6).

Both are deterministic; every script and result can be independently re-run following the attached inventory (the attached experiment-inventory file).

2.2 Machine-checked conventions

To avoid convention clashes, two points were machine-verified (attachment #25): (i) bundle-construction bin numbers and χ frequencies correspond with a ±1 shift and parity inversion (even-bin bundles → odd frequencies, and vice versa; power ratio 1.0000). The text uses the **χ-frequency convention** throughout. Fermionic = even χ frequency (≥4) = odd harmonics [7]. (ii) Bundles carry exactly one unit of η winding m=+1 by construction (η occupancy (1, 1.000)); species of arbitrary winding m are constructed by the winding shift e^{i(m−1)η}, not by projection.

2.3 Most important convention: readout level vs. dynamics level

The claims of this paper are restricted to the **readout level** — "particle classification arises as the relation between observation clocks and addresses." The **dynamics level** — "the 62 species are generated, decay, and interact as the spectrum of a single dynamics" — is outside the scope of this paper and is deferred to the next paper (the universal interaction function). The closing section (§19) fixes the requirement table for that function on the basis of the present measurements.

2.4 v2 addition: the localization prototype of the readout (used for part of Part II)

For the measurements of new pillars 8–10, we use a **localization prototype** that promotes the θ readout of the two-body system [3] (one scalar from the global spectrum) to a local field $\theta(x)$ (definitions self-contained in attachments #30–32): from band weights W_f, W_b built by the parity split (even k / odd k = the exact symmetry of the half-ring translation, scale-free) and a smooth IR roll-off $\exp(-(|k|/3)^4)$, we form $\theta(x) = \mathrm{atan2}(\sqrt{f_{loc}}, \sqrt{b_{loc}})$ and apply per-cell SO(2) rotations plus the pointwise inelastic rotation. The rotations preserve $|a|^2+|b|^2$ exactly at every cell (closure conservation; measured drift $\sim 10^{-13}$). The channel ratio $\lambda = b/a = -i$ (circular polarization) is an **exact invariant manifold** of these rotations (verified algebraically; numerical deviation 0.0). That this prototype preserves the classification of Part I is confirmed by the consistency battery of §14 (nine pillar experiments × three engines). Because localization shifts continuous quantities by a few percent, all quantitative values in Part II are flagged as prototype measurements, and the claims are restricted to scaling forms (invariance of products; order-1 statements).

3. Pillar 1: the foundation of the periodic law — the universal clock (measured)

The collective clock of the N-body condensate was scanned from N = 4 to 144 (28 points).

Measured: $\omega_{clock} = \pi/72$/step holds across the whole range — ratio 0.988±0.028 in the short window (T=4000) and 0.9992–1.0013 (**±0.1%**) in the long window (T=42000). One clock cycle = 144 steps, independent of the effective energy (N).

Fig. P1: Clock universality. $\omega/(\pi/72) = 1$ for all N.

The long-window census further shows that the only long-time stationary species in the low-energy sea (N≤16) is **the single massless ground species locked to the clock, $\rho = 1/1$** ($|\rho-1| < 9\times10^{-4}$; mass measure 10^{-11}). All massive sidebands visible in short windows are finite-lifetime transients, entrained into the clock and gone (Fig. P2). This refutation process (apparent short-window addresses and a mass law that vanish in the long window) is fully recorded in §18.

Fig. P2: Short-window sidebands (resonances) contract to 1/1 in the long window.

Hypothesis (Pillar 1): the classifier of particle species is the rational address of $U^n = I$, and the foundation is a single U^{144} clock lattice. Species are classified by "address (winding) × relation to the clock."

4. Pillar 2: the charge law — $Q_{em} = m/3$ and confinement = mod-3 readability

4.1 Multi-valuedness of charge (state side, measured)

Charged species (winding q=+1) live orders of magnitude longer than neutral transients ($\tau\approx1.3\times10^4$ collisions, 85% retention), but not forever. Their decay is not disappearance but transport to the partner (+3) by the sum-rule walk $m^* = 2m_B - m_s$ [4] (Fig. P3). Furthermore, isolated pure winding species **self-replicate exactly for any m** (retention 1.000; τ numerically ∞), and even the sea-free mixture {+1,+3} does not leak — **the walk requires the m=0 sea as its driver** (the leak targets {0,−1,+2}/{±4} agree completely with products of the sum rule).

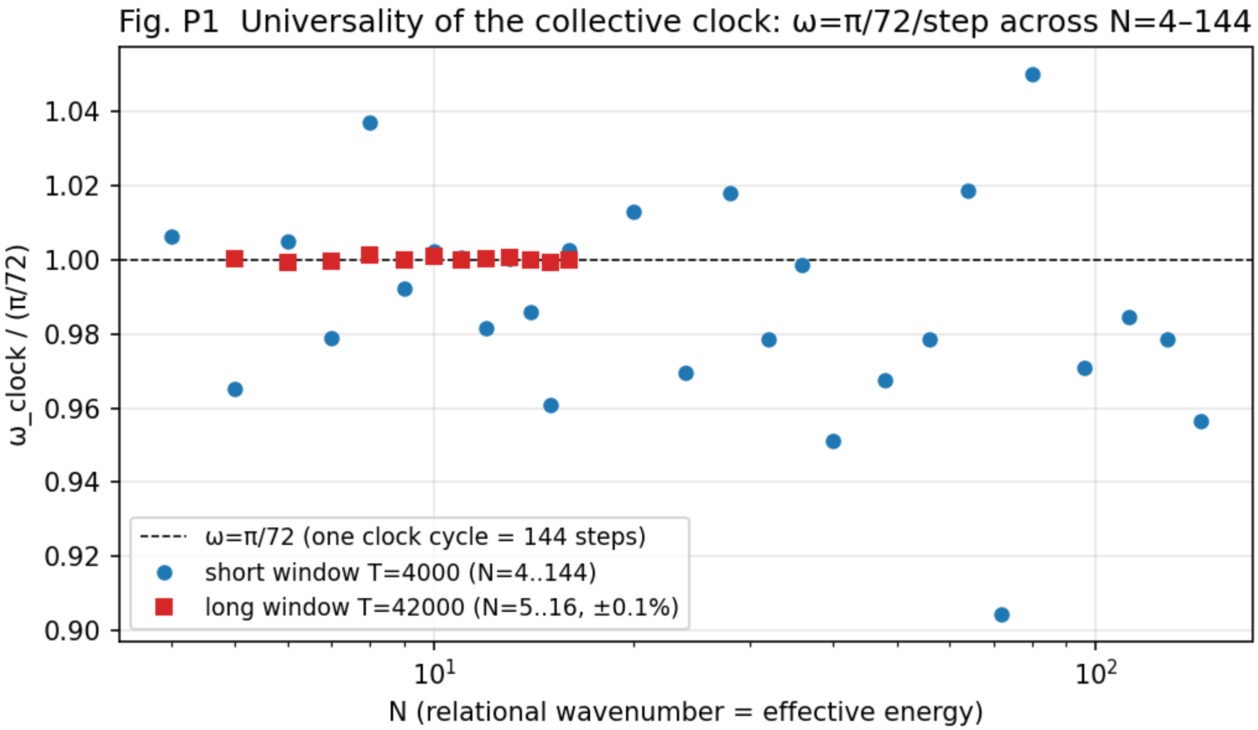

Figure 1: Fig. P1

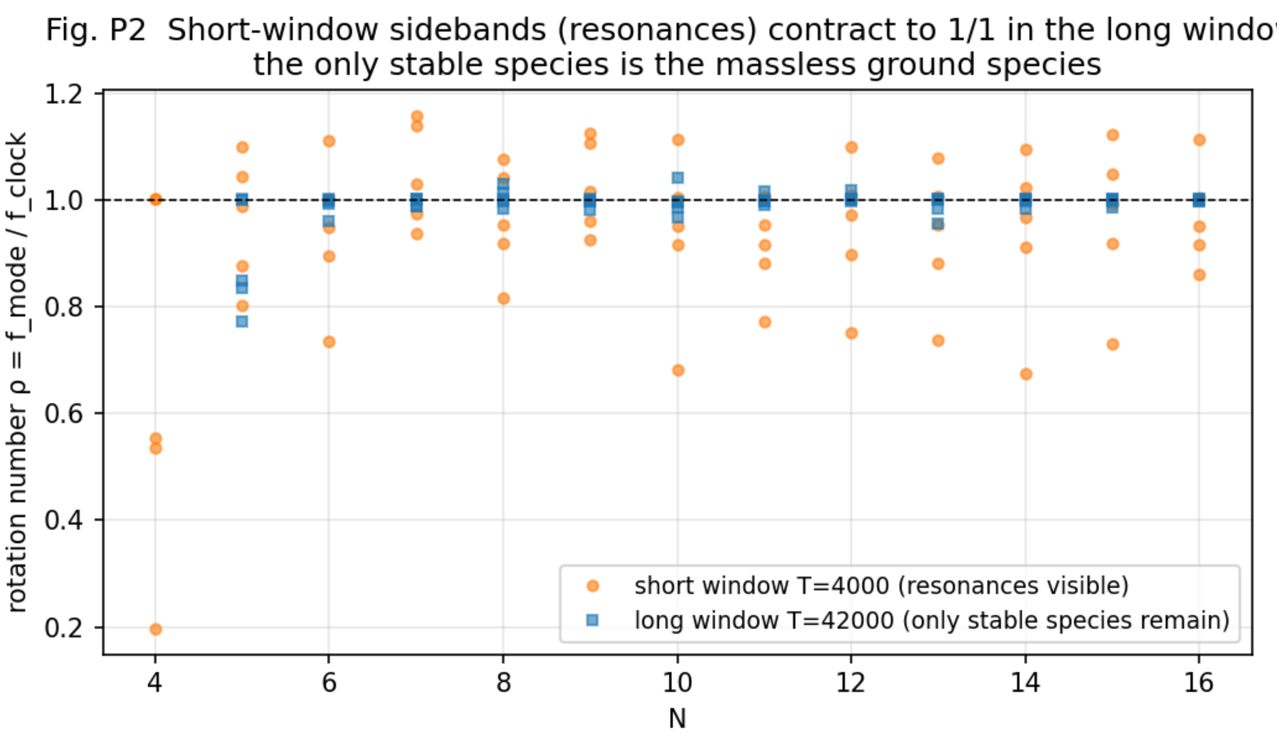

Figure 2: Fig. P2

The “charge does not stabilize at ±1” of the previous paper [5] §9.5 is a direct consequence of this sea-driven walk.

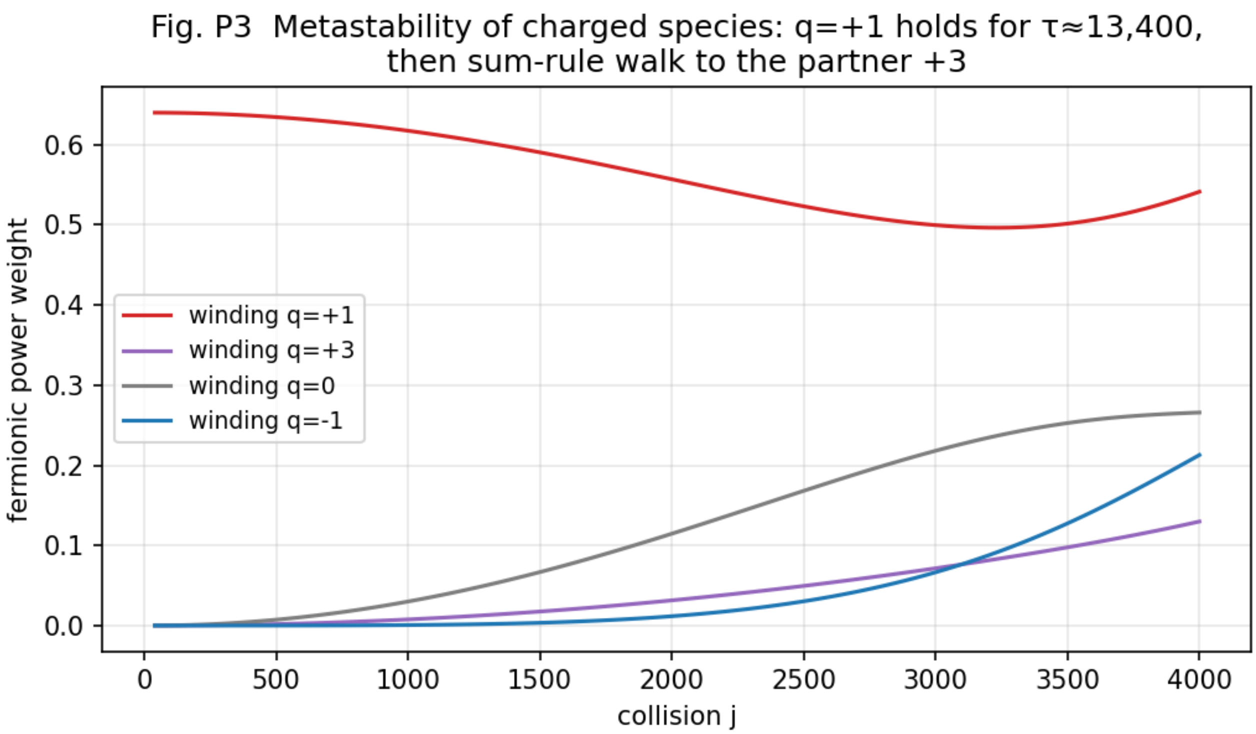

Figure 3: Fig. P3

Fig. P3: Metastability of charged species and the sum-rule walk (+1 → +3).

4.2 The conservation law of winding charge is cyclic

The true conservation law of winding charge is not integer conservation but **cyclic conservation mod ne** (the η register order; 16 in this system). The pointwise interaction is a cyclic convolution of the η spectrum (in the continuum limit $\delta Q = R\oint\partial\_\eta(s^2) = 0$), and conservation of the integer charge Q_wind is merely the in-band special case (measured exactly, CV≈0, in the charged census construction, Fig. P6a). The apparent breaking is the fold-back of the doubling walk reaching Nyquist (37% edge-power accumulation; correlation −0.93; Fig. P6b). On a finite register the doubling ladder neutralizes in 4 steps: 1→2→4→8→0.

Fig. P6: Cyclic conservation and Nyquist fold-back.

4.3 Readout rectification (readout side, measured)

Against all products of the sea-driven walk (the doubling/inversion ladder 2^b·(±1)), **only the observation clock J=3 reads all charged content at |q|=1** (concentration 1.000 — in a universe seeded with +1 as well as one seeded with +2). J=4 erases charge; J=5, 6 scatter it into many values (Fig. P4). The algebraic basis is 2^b mod 3 = ±1. That the observation clock has denominator 3 rests on prior measurement [6][7] — the locking tongue widths are dominated by denominator 3 with a width ratio >461. The value of the elementary charge itself connects to the published measurement [6] as the rational address $\sin^2(23\pi/124) = \sqrt{4\pi\alpha}$ (agreement $16/10^8$).

Fig. P4: Readout rectification. Only the divide-by-3 clock reads all charged content at |q|=1.

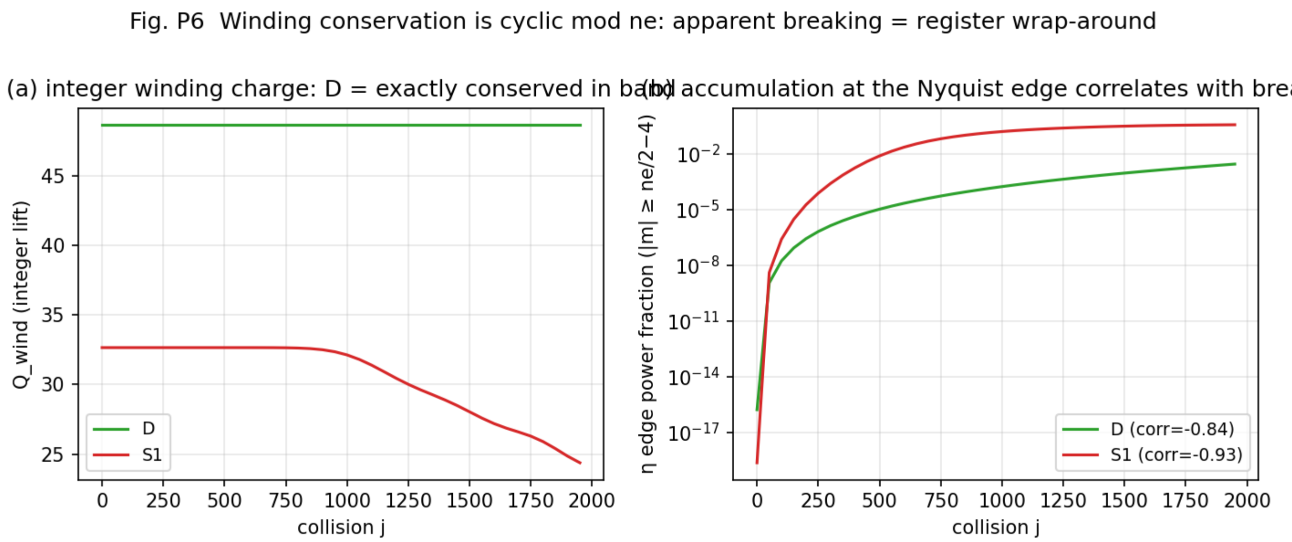

Figure 4: Fig. P6

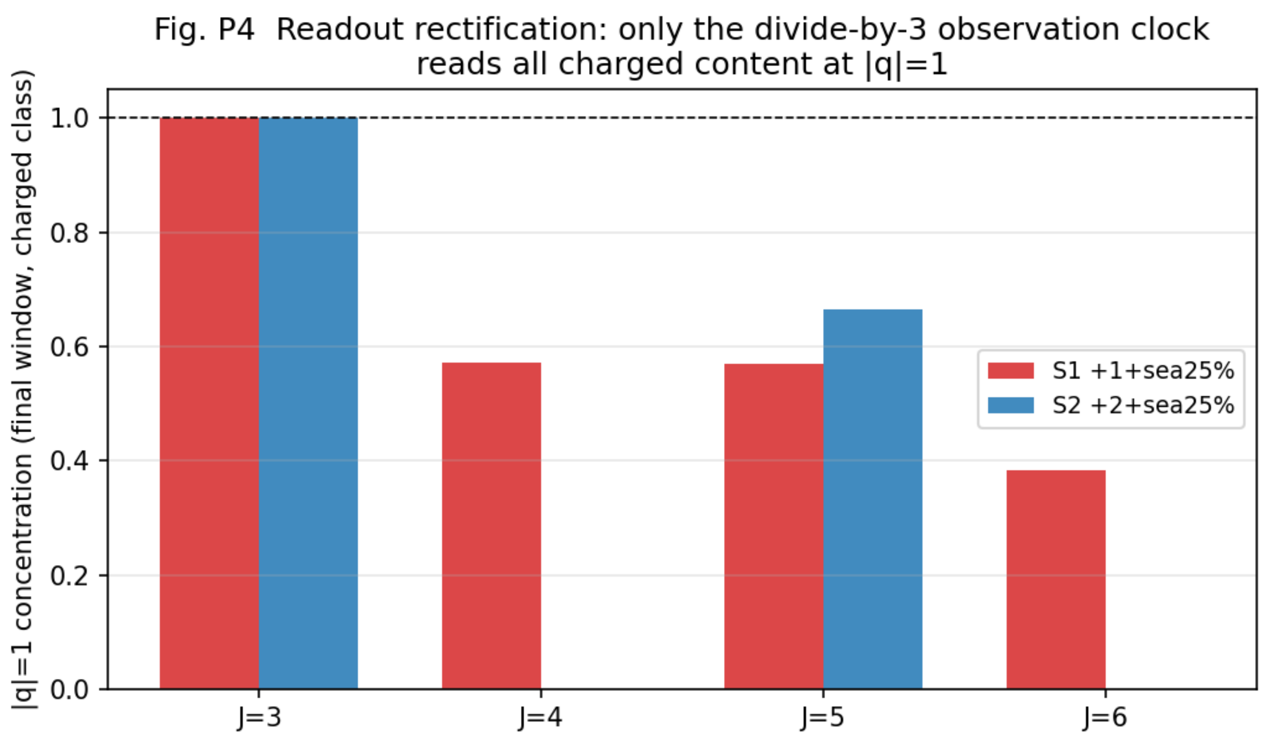

Figure 5: Fig. P4

The bookkeeping also closes: in conserving constructions, $\Delta Q3 = -3\Delta W$ holds to 7×10^{-10} (Q3 = net charge readable mod 3; W = the ledger of charge hidden in mod-3-neutral composites). **Readable charge does not disappear — its decrease equals exactly what is carried into composites that the denominator-3 clock reads as neutral** (Fig. P5).

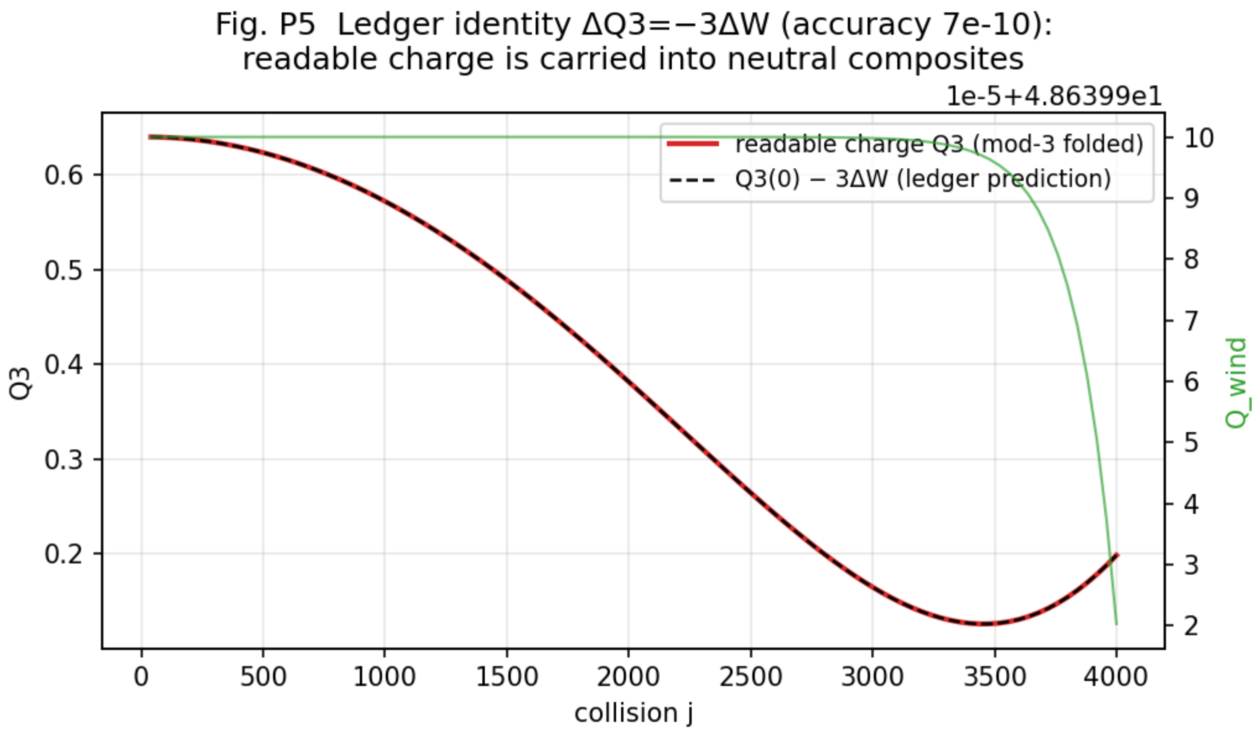

Figure 6: Fig. P5

Fig. P5: The ledger identity $\Delta Q3 = -3\Delta W$.

4.4 Hypothesis (Pillar 2): Q_em = m/3 and confinement

We propose the charge law binding the above: **charge is Q_em = m/3, and the dominant observation clock (denominator 3) reads "3 raw windings = one elementary charge."** Consequences:

- u type m=+2 → +2/3; d type m=−1 → −1/3; electron m=−3 → −1; ν m=0 → 0.
- **Confinement = mod-3 readability**: the observer's space is the Z_3 quotient of the η circle, and only species single-valued on it (m≡0 mod 3) can be read freely. Quarks (m≢0) are unreadable alone — **1/3 charge and confinement are two faces of one fact** and need no separate explanations.
- Check: proton uud = 2+2−1 = +3 → Q=+1. Neutron udd = 2−1−1 = 0 → Q=0. $\pi^+(u\bar{d})$ = 2+1 = +3 → +1. **Integer charges of all hadrons follow automatically.**

Confinement test (measured): quark type (m=+2) + sea starts from readable fraction f_read = 0.0000 and becomes readable exactly to the extent that the walk forms m≡0 composites (hadron analogues) (→0.2504 over 4000 collisions). Electron type (m=+3) + sea starts from f_read = 1.0000 (free from the start). **The isolated quark-type system stays at f_read = 0.0000 even after 4000 collisions** — "a single quark cannot be observed" is realized as a consequence of readout (Fig. P9).

Fig. P9: The confinement = mod-3 readability test.

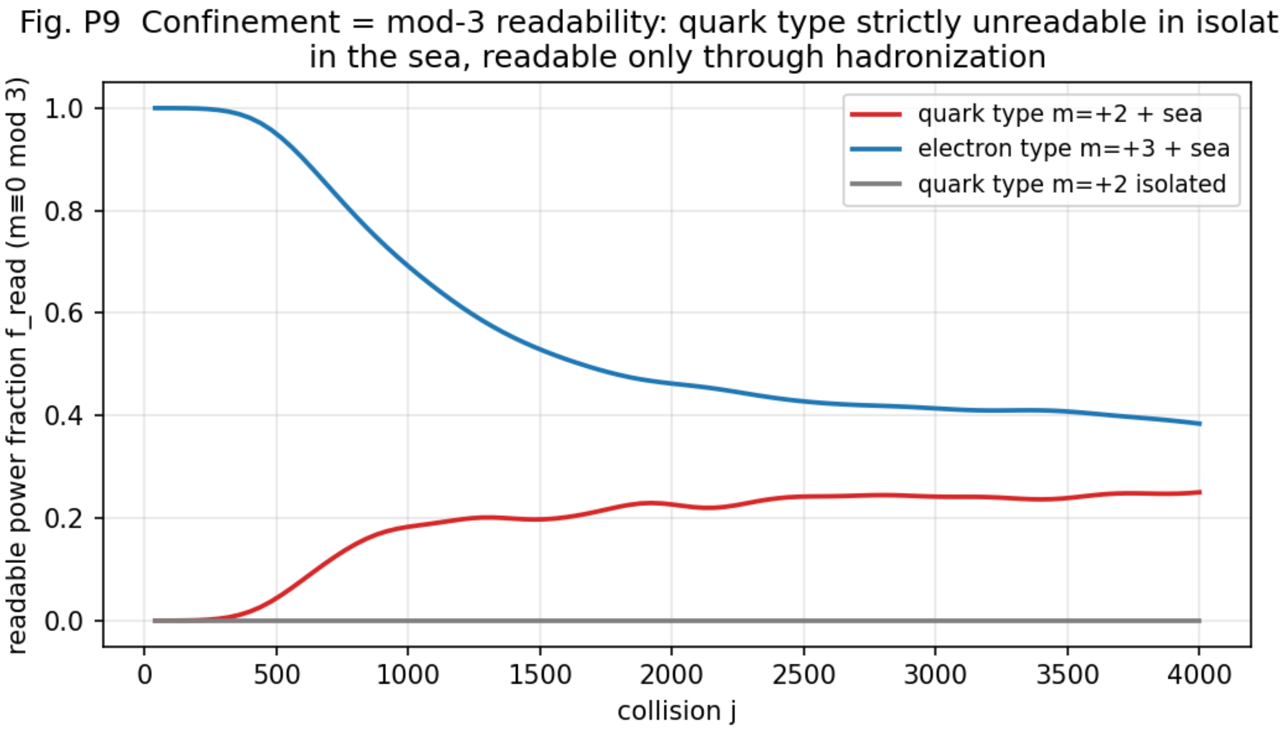

Figure 7: Fig. P9

Correspondence with the standard theory: the structure is isomorphic to the Z_3 superselection of the SU(3) center — "free states are triality 0 only" [10][11]. But whereas in the standard theory Z_3 enters as an axiom on the dynamics side (the center of the gauge group), here mod 3 comes out of the **readout side**, as the dominant denominator of the observation clock (measured) — one of the novelties of this paper (§15).

Two precise distinctions: (i) this section contains two logics that must be kept apart — "the denominator-3 clock rectifies the charged walk to ±1" is measurement; "identifying physical charge with m/3" is hypothesis. To promote the identification to a law requires the dynamical test of **response** — whether the interaction strength Γ_int of species with different m scales as m/3 (or $(m/3)^2$) (§19). (ii) The confinement here is **readout-level confinement** (unreadability in isolation); the standard notion also contains growing separation energy, flux tubes, and asymptotic freedom, none of which are derived here. Precisely, this is a **readout precursor structure of confinement**; the test of whether separating two unreadable species costs energy increasing with distance (§19) is the condition for promotion to dynamical confinement.

5. Pillar 3: the statistics law — the Z_2 clock cover

5.1 Statistics does not live in spatial rotation (measured)

Global rotation of the channel doublet is an exact symmetry of the dynamics (control runs flat at machine zero for all observables; rotation invariance of $\mathrm{Im}(\bar{b}a)$ verified analytically), and the spatial rotation charge is $\ell=\pm1$ for all observed modes (after calibration, deviations within ±0.01 at N=6, 8) — neither half-integers nor $\ell=2$ appear in occupied modes. The two-valuedness of statistics lives neither on the channel side nor on the spatial side.

5.2 Statistics lives in the double cover of the clock (measured)

Tracking the state phase of a species along the recurrence sequence of the observables (the autocorrelation peak sequence of the observable field $s = \mathrm{Im}(\bar{b}a)$), **the phase of charged species is strictly quantized to {0, π} at every recurrence point** ($\Phi/\pi = \pm 0.999$ or $+0.002$; 3-digit precision; no continuous values) — the state commutes between the +1 and −1 sheets (Fig. P8). Neutral non-projected bundles show continuous phases; the presence or absence of quantization separates the species.

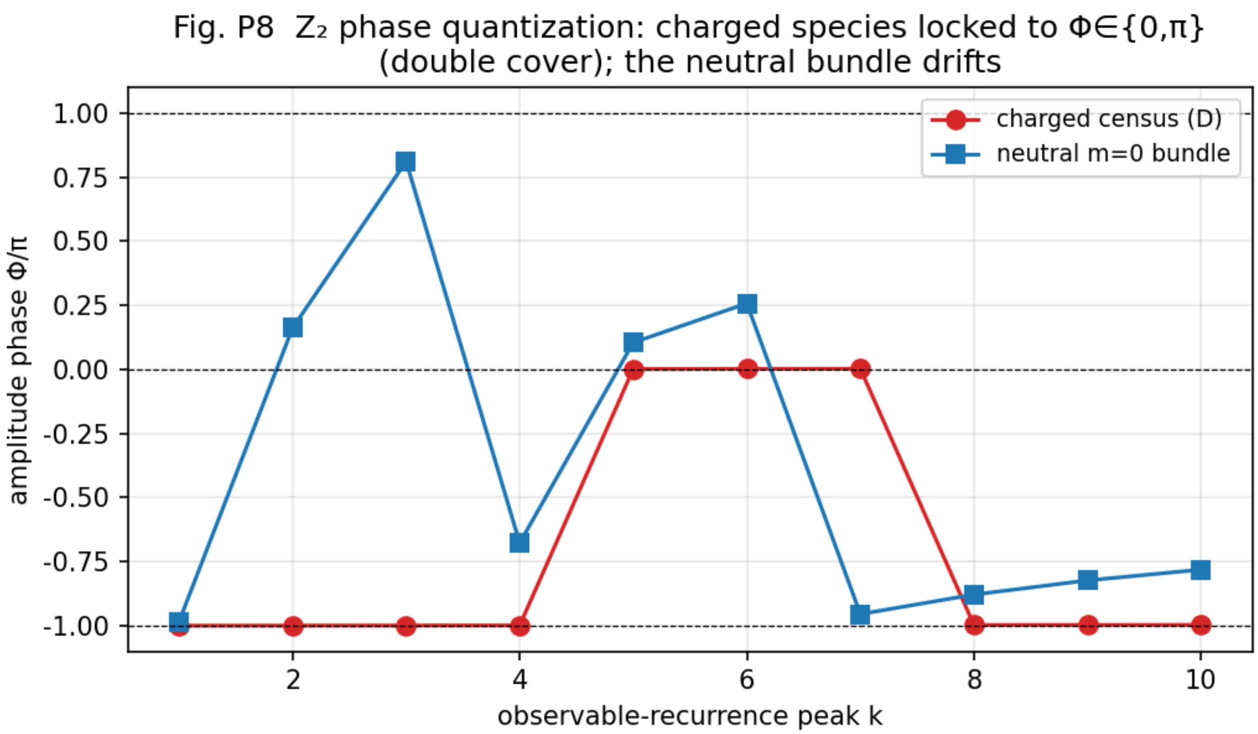

Figure 8: Fig. P8

Fig. P8: Z_2 phase quantization. Charged species: $\Phi \in \{0,\pi\}$; the neutral bundle drifts.

The cross-table experiment (4 cells: χ parity × winding {1,2}) shows that **the covering degree depends only on χ parity and not on the winding (charge)**: fermionic species (even χ frequency = odd harmonics) = covering degree 2 (Z_2, both sheets); bosonic species = covering degree 1. Moreover a pure m=0 species in the fermionic band (winding-shift construction; η occupancy (0, 1.000)) also shows covering degree 2 with Qz2 = 1.00 — **the row of the neutral fermion (neutrino) is established** (Fig. P10).

Fig. P10: The neutrino row and the spin-statistics correspondence.

5.3 Hypothesis (Pillar 3) and prior contrast

We propose **fermion/boson distinction = clock covering degree** (1 = bosonic / 2 = fermionic). This corresponds to the structure by which Finkelstein–Rubinstein [12] derived the spin-statistics of solitons from the double cover of configuration space — but differs in that **the double cover lives not in space (rotation, exchange) but in the clock (state period = observation period × 2)**. The structure connects to the prior measurement [6][7] of the order-248/observation-124 fidelity (no recurrence visible on the F_{124} side; $F_{248} = 1$). (v2 addition: the double cover also has a spatial embodiment — not as rotation but as **antipodal structure**; see the spatial double-cover theorem of §10.)

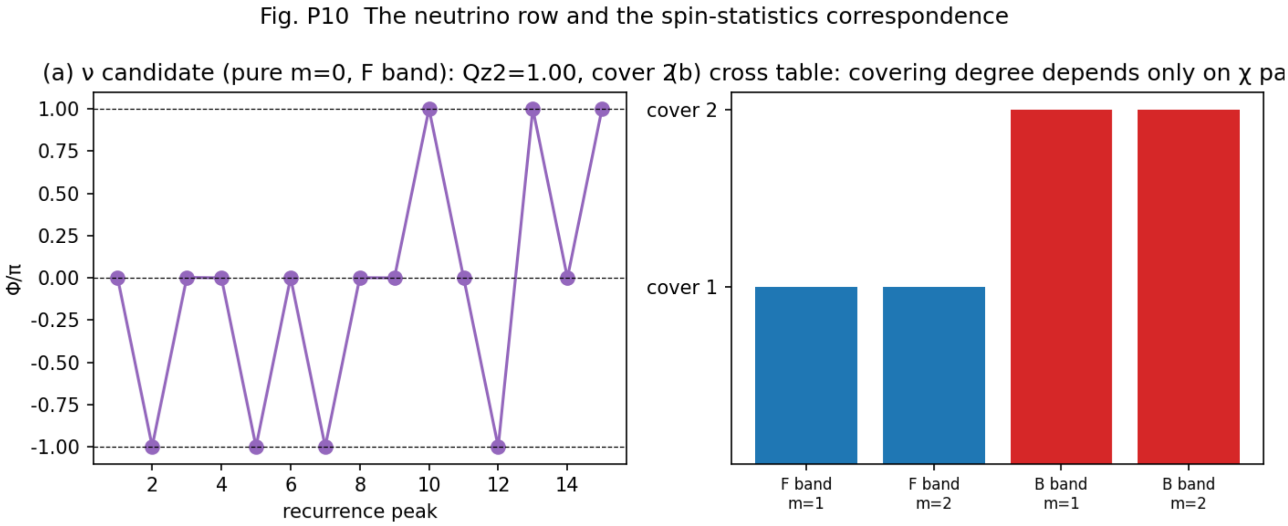

Figure 9: Fig. P10

6. Pillar 4: the mass law — mass is a relation to the sea

6.1 Equivalence theorems (three, measured)

- **Winding-shift equivalence**: $e^{im\eta}$ is an exact symmetry of the dynamics ($\bar{b}a$ invariant); isolated pure-address species are all unitarily equivalent — measured: mass^2, polarization, s_z agree at machine precision for all addresses. **Mass and spin are not attributes of an isolated address.**
- **Divisor-class theorem**: in the sea, properties differentiate — but **only down to the divisor class gcd(m, 16)** (mass^2 of odd {1,3,5,7} matches to 5 digits; {2,6} match; {4} distinct; Fig. P7). The reason is exact: the unit automorphisms $m \to um$ of Z_{16} fix the sea ($m=0$), so sea coupling cannot distinguish unit addresses in principle. Consequence: **the distinction between ±1 and ±5, ±7 does not exist in the dynamics; only the mod-3 readout breaks the unit class** — independent support for Pillar 2 (readout rectification).
- **χ-band translation symmetry**: placing the same winding species on χ bands (10,12,14)/(30,32,34)/(50,52,54) leaves mass^2 identical to 5 digits — band position does not differentiate species either (§16, weakness 1).

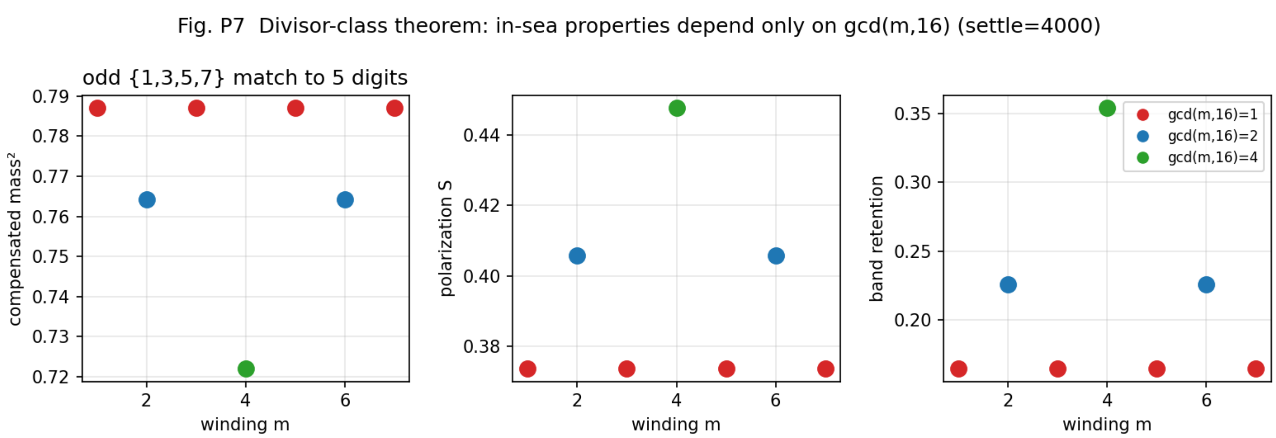

Figure 10: Fig. P7

Fig. P7: The divisor-class theorem. In-sea properties depend only on gcd(m,16).

6.2 Hypothesis (Pillar 4)

Mass is a relational quantity with the sea (vacuum). Only the species locked to the clock (ρ = 1/1) is massless and stable (the photonic ground species; mass measure 10^{-11}, measured); content detuned from the clock carries mass as resonance; and the sea differentiates the individuality of species (lifetime, polarization, mass) at the level of divisor classes. The view of the vacuum-as-sea as a co-determiner of particle properties resonates with the lineage of emergent gauge and emergent fermions [13] (string-net condensation; superfluid ^{3}He), but this paper differs in having measured the sea dependence of properties in the form of equivalence theorems. (v2 addition: the readout embodiment of mass is the mean of the local clock field, ⟨ω⟩ — see §11. The "mass → proper clock" framework originates in the previous paper [5].)

7. Pillar 5: the periodic table of waves — the two-sided table and the 62-species assignment

7.1 The two-sided table

- **Table A (isolated stable elements)**: pure address species on the U^144 clock lattice. All exactly stable by self-replication (measured). Algebraic classifiers: readable charge (mod 3), doubling orbit, neutralization steps, η parity. A second parity structure: odd addresses resist doubling neutralization maximally (4 steps); even ones fall fast (±4 in 2 steps; +8 in 1).
- **Table B (in-sea lifetimes)**: sea coupling selects species — charged means long-lived ($\tau \sim 10^4$), neutral transients short-lived ($\tau \sim 10^{2-3}$), composites neutralize. **Actually observed "particles" are this metastable hierarchy.**

The native periodic table (model-intrinsic, the measured primary table): we first place the table written purely in the classifiers of this paper, without Standard-Model names. The Standard-Model assignment (§7.2) is a hypothesis on top of this table.

Raw winding class	mod-3 readable	Neutralization steps	Clock cover (F/B band)	In-sea mass² (divisor class)	Isolated stability	In-sea lifetime
m=0	0 (free, neutral)	0	2 / 1	— (sea, ground species)	stable	ground species stable
odd (units) {±1,±3,±5,±7}	±1 or 0	4 (max)	2 / 1	0.787 (5-digit match)	exactly stable	$\tau \sim 10^4$ (charged)
{±2,±6}	∓1 or 0	3	2 / 1	0.764	exactly stable	retention 0.226
{±4}	±1	2	2 / 1	0.722	exactly stable	retention 0.354 (longest)
{+8}	−1	1 (shortest)	—	unmeasured	—	unmeasured

The covering degree is decided by χ parity, not the raw winding class (cross-table measurement), hence an independent column.

7.2 The assignment of the 62 Standard-Model species

Confidence tags: **E** = measured anchor (directly supported by measurements here or already published) / **S** = structural correspondence (mechanism matches; quantitative check pending) / **H** = hypothesis (target of future tests).

Particle	States	Winding m ($Q_{em}=m/3$)	Cover	Conf.	Notes
u, c, t	18	+2 (+2/3)	2	**E**	$m\neq0$ → confinement (Fig. P9 measured). Color = mod-3-unreadable raw-winding residue (ledger W)
d, s, b	18	−1 (−1/3)	2	**E**	same
e, μ, τ	6	−3 (−1)	2	**E**	free ($m\equiv0$). Elementary-charge address $\sin^2(23\pi/124)$ [6]
ν ×3	6	0 (0)	2	**E**	neutral fermion (established by §5.2)
γ	1	0	1	**E**	ground species ρ=1/1 (massless, stable, measured) = vacuum fixed point (§9)
g	8	color pair (mod-3 neutral, raw winding nonzero)	1	S	non-singlet color → unreadable alone = confinement. Octet generation not derived
W±, Z	3	±3, 0	1	S	t winding (detuning of its own clock) → massive (consistent with entrainment measurements; dynamics untested)
H	1	0	1	S	collective mode of the sea (condensate). $\ell=0$

Particle	States	Winding m (Q_em=m/3)	Cover	Conf.	Notes
G	1	0	1	H	only as a composite of two ℓ=1 quanta (§8.4 prediction). ℓ=±2
(generation axis)	—	—	—	H	v2 candidate: generations = discrete series of phase-lock levels (§11)

Total **62 species** (36 quarks + 12 leptons + 8 g + γ + 2 W + Z + H + G). Antiparticles are m→−m (charge conjugation = η inversion; pair creation measured [3]). The confidence of each row (measured anchor / structural / hypothesis) is made explicit in the attached v2 periodic-table visual (Fig. T, Japanese and English). **Scope of the confidence tags**: the row confidence applies to the primary attributes (charge identification, statistics cover, confinement); **the generation assignment (e.g., mapping all three generations u, c, t to the single address m=+2) and quantitative masses are H for all rows** (the differentiation axis of generations is the candidate of §11.3; verification program §19.2-6).

Fig. T: The periodic table of waves v2 (the one-line law, the native table, the 62-species assignment, the mass-lifetime law, and the hierarchy of existence).

7.3 The relation of the two finite structures (an open structural problem)

Two finite structures are measured in this paper: the universal clock U^144 (§3) and the winding register Z_{16} (§4.2). The factorizations are $144 = 2^4 \cdot 3^2$, $16 = 2^4$, $144/16 = 9 = 3^2$, and the classifiers of this paper correspond to these factors — **2 = statistics cover (§5), 3 = charge readability (§4), 16 = raw winding cycle (§4.2)**. Whether this coincidence is accidental, or a resonance structure in which both follow from the update rule (a resonance classification of clock order × internal winding order), is the largest open structural problem of this paper. It is registered in §19 as a task to be tested **as the recurrence structure of an operator**, not as numerology: bring each readout operation into the same operator representation and measure the order ord(U|_{H_i}) per subspace — the true periodic law is possibly the family of "orders that a single operator has on different readout subspaces." The real periodic law of the "periodic table" probably lives here.

8. Explanatory power — connections to existing problems

8.1 The open problem of the previous paper, "charge unstable at ±1", resolved

The previous paper [5] §9.5 recorded as an open problem that "charge (winding number) does not stabilize at ±1 but distributes over several values." Under the present hypothesis this is no contradiction: on the **state side**, the sea-driven walk (measured, §4.1) keeps dispersing the winding; on the **readout side**, the denominator-3 observation clock folds all its products back to |q|=1 (measured, §4.3). "Multi-valuedness of charge" and "universality of the elementary charge" coexist without contradiction — the open problem turns into a prediction of the hypothesis.

8.2 The 1/3-charge problem of quarks

Under Q_em = m/3, fractional charge (why 1/3) and confinement (why never seen alone) are a single mechanism, mod-3 readability (§4.4; measured, Fig. P9).

The Periodic Table of Waves, v2

particle species = state-side address × observation clock × relation to the sea (v2: mass, lifetime, hierarchy unified)

The one-line law (the trunk of v2): $r = \sin^2\theta = P_{odd} / (P_{odd} + P_{even})$ — collision rate = fraction of odd-harmonic (fermionic-band) power

Table 1 Native periodic table (model-intrinsic, measured) — the primary table without Standard-Model names

Raw winding class	Readable charge (mod 3)	Steps to neutralize	Clock covering (F/B band)	In-sea mass² (divisor class)	Isolated stability	In-sea lifetime
m=0 (sea, ground species)	0 (neutral, free)	0	2 / 1	—	stable	ground species stable (lightlike)
odd class {±1,±3,±5,±7}	±1 or 0	4 (max)	2 / 1	0.787 (5-digit match)	exactly stable	$\tau \approx 10^4$ collisions (charged)
{±2, ±6}	∓1 or 0	3	2 / 1	0.764	exactly stable	retention 0.226
{±4}	±1	2	2 / 1	0.722	exactly stable	retention 0.354 (longest)
{+8}	−1	1 (shortest)	—	unmeasured	—	unmeasured

Table 2 Assignment of the 62 Standard-Model species (hypothesis) — confidence: E=measured anchor / S=structural / H=hypothesis

Particle	States	Winding m (charge Q=m/3)	Statistics (cover)	Conf.	Notes
u, c, t	18	+2 (+2/3)	fermion (2)	E	confinement = mod-3 unreadability; color = hidden residue
d, s, b	18	−1 (−1/3)	fermion (2)	E	same as above
e, μ, τ	6	−3 (−1)	fermion (2)	E	free; elementary-charge address $\sin^2(23\pi/124)$
ν×3	6	0 (0)	fermion (2)	E	neutral fermion established by measurement
photon γ	1	0	boson (1)	E	ground species ρ=1/1, massless, stable = vacuum fixed point
gluons g	8	color pair (mod-3 neutral)	boson (1)	S	non-singlet color → unreadable alone
W±, Z	3	±3, 0	boson (1)	S	t-winding (clock detuning) → massive
Higgs H	1	0	boson (1)	S	collective mode of the sea (condensate)
graviton G	1	0	boson (1)	H	only as a composite of two ℓ=1 quanta (prediction)
(generation axis)	—	—	—	H	v2 candidate: generations = phase-lock levels (full/partial/marginal)

Mass-lifetime law (v2, prototype): mass = ⟨ω⟩ (mean clock rate) / lifetime = $1/\sigma_\omega$ (inverse linewidth) / $\tau \cdot \sigma_\omega \approx$ const (CV 30%) ⇒ heavier ⇒ shorter-lived (generation phenomenology automatic)

Table 3 The hierarchy of existence (new in v2) — seating chart of the baryon hierarchy, mass, and lifetime

Level	Organizing principle	Charge	Statistics	Mass	Lifetime / decay	Conf.
sea (vacuum)	pure bosonic band (P_{odd} = 0)	0	—	does not tick (exact fixed point, r=0)	∞	E
elementary (62)	pure parity = antipodal two-point structure (no point localization)	Q=m/3	definite (cover 1/2)	⟨ω⟩	$1/\sigma_\omega$	S
hadrons / baryons	ω-locked class with Σm≡0 (mod 3) p=uud:+3→+1 / n=udd:0→0	integer (automatic)	composite	⟨ω⟩ of the class	splitting = class bifurcation $t_{split} = \pi/\Delta\omega$	S
classical bodies	equal parity mixture (all harmonics, point-localized)	integer	none (mixed)	Σ⟨ω⟩ (gravity source)	macroscopic	S

62 in total = 36 quarks + 12 leptons + 8 gluons + γ + 2 W + Z + H + G (antiparticles: winding m→−m)

charge = reading of the divide-by-3 clock / statistics = double cover of the clock (spatially: antipodal two-point structure) / mass & lifetime = mean and width of the clock field ω(x) / splitting = bifurcation of ω-locked classes (natural π criterion)

The Periodic Table of Waves v2 (doi:10.5281/zenodo.21822358) confidence: E=measured anchor / S=structural / H=hypothesis

Figure 11: Fig. T

8.3 Antimatter

Charge conjugation $m \to -m$ is a symmetry of the spectrum, and pair creation is a necessity of the walk (published measurement [3]).

8.4 The weakness of gravity

The rotation-generator spectrum of the condensate has no linear $\ell=2$ frame (calibration ratio at most 1.000 exactly; measured). Spin 2 can exist only as a **composite** of two $\ell=1$ quanta, and indeed the spectrum of the squared readout carries a coherent 2ω quadrupole line at all N (ratio 2±6%). The measurements end here. Beyond this we state a conjecture: if it is a second-order composite, its coupling is expected to suffer square suppression as a product of first-order quantities (the connection to an $\alpha_G=(m/M)^2$-type gravitational coupling [15] is untested — a prediction); if it passes, **the reason gravity is weak and the reason it is hard to test become one structure**. This is a candidate appearance, from the state-space side, of a structure isomorphic to gravity amplitude = gauge amplitude^2 (double copy [16]).

8.5 v2 addition: correlation of the mass and lifetime hierarchies

In the charged-lepton series e, μ, τ of the Standard Model, a striking pattern holds: as the generation rises and the mass grows, the lifetime shortens (for quarks the very concept of a free-particle lifetime is not simple because of confinement). Conventionally this correlation is computed case by case from decay phase space. New pillar 8 (§11) offers a structural candidate: if mass and linewidth (inverse lifetime) are **the mean and the width of one and the same clock-field distribution**, their correlation is a consequence of structure, not of computation.

8.6 v2 addition: why elementary particles are not points

New pillar 7 (§10) gives as a geometric theorem: "species with definite statistics carry an antipodal two-point structure and do not localize at a point." It is the in-model counterpart of the circumstance that in quantum field theory particles are modes, not points. Conversely, a body localized at a point necessarily mixes both parities and has no statistics — that point particles have no statistics and that statistical species are not points are two faces of the same geometry.

8.7 v2 addition (conjecture, confidence H): a hint that vacuum energy does not gravitate

A uniform sea contributes only a uniform offset to the local clock field and nothing to its **gradient**. If gravity is read as the gradient of the clock field, vacuum energy may be absorbed into a uniform gauge shift and fail to gravitate. This is a conjecture (confidence H), registered as a target of the next paper (the gravity readout).

Part II — The new findings of v2: unification by the clock field ω(x) (new pillars 6–10)

Of the measurements in Part II, §9(A) is on the published system itself (global θ, [3]); §§11–13 are on the localization prototype of the readout (§2.4). The legitimacy of the prototype rests on the consistency battery of §14. Quantitative values are flagged as prototype measurements, and claims are restricted to scaling forms.

9. New pillar 6: the one-line law — r = P_odd/(P_odd+P_even)

9.1 The law

Exactly from the definition of the θ readout of the two-body system [3], the reflection rate per interaction is

$r = \sin^2\theta$ = P_odd / (P_odd + P_even)

(P_odd = the power of the fermionic band = the intrinsic odd harmonics; P_even = the rest). **The reflection rate per interaction — how much is exchanged — is decided by the odd-harmonic fraction alone** (r is a rate, not a probability; the interaction acts at all times). Even harmonics enter the rotated side but appear only in the denominator of what decides the rotation. The response signs are opposite: $\partial r/\partial P\_odd = P\_even/P^2 > 0$, $\partial r/\partial P\_even = -P\_odd/P^2 < 0$ — even at equal amounts the roles remain distinct (the analogue of the ± of charge).

9.2 Consequences (all measured)

1. **The vacuum fixed point**: the pure bosonic sea (P_odd=0) is exactly stationary at r=0 — **light does not collide with light, and the vacuum does not tick** (std of τ_t $\sim 10^{-16}$ in control runs). It is the dynamical embodiment of the γ row of Pillar 5 (ground species; massless; stable). **The two clocks distinguished (important)**: “the vacuum does not tick” means that the **local reflection clock of the two-body system** (the local readout τ_t) does not advance; it does not mean that the **collective phase clock of the N-body condensate**, ω_clock = π/72 (Pillar 1), does not exist. The relation between the two clocks — the local reflection clock and the collective phase clock — is registered as an open problem in §16 (if unified, Pillar 1 and new pillar 6 connect).
2. **The candidate source that drives the clock and mass readouts = odd-band power**: the source that moves the clock is odd-harmonic content only. The role asymmetry — matter = the side that causes interaction; light = the side that does not — follows from the formula. (What is exact is P_odd → r; P_odd → mass passes through the connection hypotheses of §11 — r → ω(x) → ⟨ω⟩.)
3. **The necessary condition for the Z_2 classification to have dynamical meaning (within this model)**: if r were parity-symmetric, no readout of this model could distinguish the fermionic and bosonic sectors, and the Z_2 classification would degenerate into nomenclature. The necessary condition for the statistics classification to make a readable difference in this model is this asymmetric coupling (we do not claim a necessary-and-sufficient condition for fermi/bose statistics in general).

9.3 The asymmetry, measured (attachment #29)

An equal-amplitude initial condition — a bundle tuned by bisection to fermionic fraction f_0 = 0.500000 exactly — evolved for 3000 collisions in the published system (global θ) **flows out to f* = 0.4936 ± 0.0012**. f = 0.5 is not a fixed point = the parity coupling is indeed asymmetric (Fig. V2a). The breaking is small, about 1% — the dynamics is nearly parity-symmetric, with a slight breaking (near-symmetry; the near-degeneracy of the two sectors is registered as an observation to track).

Fig. V2a: Equal-amplitude f = 0.500 is not a fixed point (asymmetry of the parity coupling).

10. New pillar 7: the spatial double-cover theorem

10.1 The even/odd distinction is a transformation response

The even/odd distinction is not an absolute label but the **Z_2 representation of the half-translation T: x → x + n/2 of the fundamental wavelength λ_0** (even = +1 representation; odd = −1). The projectors $P\pm = (1\pm T)/2$ separate exactly regardless of amplitudes — measured (attachment #29B): applied to the full-harmonic, center-phased ladder (an equal mixture, even

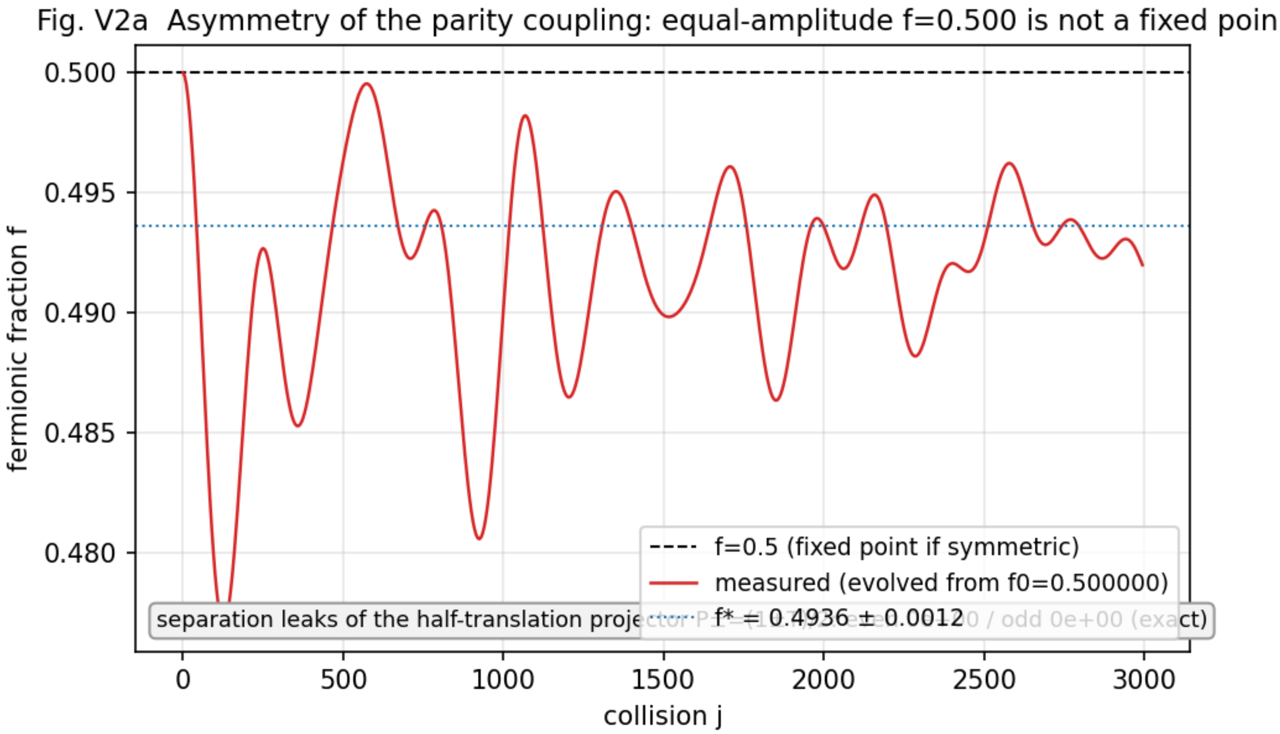

Figure 12: Fig. V2a

49% / odd 51%), the odd-k leak of the even part and the even-k leak of the odd part are both **0.0** (machine precision). “If both are present at equal amplitude they cannot be distinguished” does not hold — the discriminator is not amplitude but transformation response.

10.2 The theorem (constructive)

As direct consequences of the half-translation representation:

1. **Pure-parity species (species of definite statistics) cannot localize at a single point and necessarily carry an antipodal two-point structure** (even = symmetric image; odd = antisymmetric image). The double cover of the clock (Pillar 3) is realized as spatial structure.
2. **A point-localized body requires parity mixing and carries no statistics** (composites; classical bodies; see the hierarchy of §13). Precisely: a strictly point (delta) localized state requires **equal norms** of even and odd parts; finite-width localization admits width-dependent deviations (the measured full-harmonic ladder is even 49% / odd 51%; attachment #29B).

10.3 Position

Pillar 3 said “the double cover lives in the clock.” New pillar 7 reinforces it: the double cover also has a spatial embodiment — not as rotation (Finkelstein–Rubinstein type [12]) but as **antipodal structure**. The covering degree of the clock (dynamical) and the antipodal structure (geometric) are two faces of the same Z_2.

11. New pillar 8: unification of mass, lifetime, and generations — the mean and width of the clock field

11.1 Definitions and prediction

Take the distribution of the local clock field $\omega(x)$ (the phase advance per step; the readout $\tau_t(x)$) on the support of a species:

- **mass = ⟨ω⟩** (the mean clock rate on the support)
- **lifetime = 1/σ_ω** (the inverse linewidth; σ_ω = the spatial std of the clock rate on the support)

Prediction: **the linewidth-lifetime relation τ_coh · σ_ω ≈ const** — the dynamical version, as statistics of the clock field, of the energy-time uncertainty (the Weisskopf–Wigner linewidth-lifetime relation [23]).

11.2 Measured (attachment #30; localization prototype)

Over 8 configurations (composition f, amplitude, and harmonic width varied independently; σ_ω swept over a 6× range):

1. τ_{coh} is monotonically anti-correlated with σ_ω (σ_ω 0.044→0.247 as τ 420→99).
2. **The product τ_coh·σ_ω = 32.1, CV 29.7%** (7 uncensored) — order-1 invariance over the 6× range (Fig. V2b).
3. **Correlation of mass ⟨ω⟩ and linewidth σ_ω: +0.904** — the heavier, the broader the linewidth, the shorter-lived.

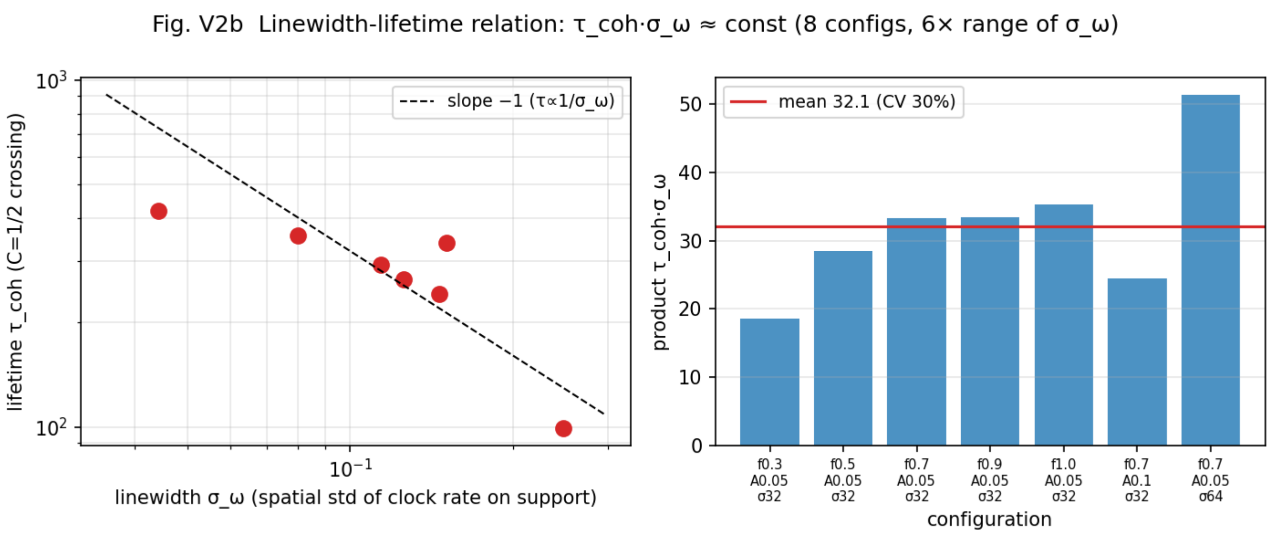

Figure 13: Fig. V2b

Fig. V2b: The linewidth-lifetime relation τ·σ_ω ≈ const, and the positive mass-linewidth correlation.

11.3 The new candidate for generations

In v1, the hypothesis "generation = χ-band position" had been refuted (χ-band translation symmetry; §6.1). New pillar 8 supplies a replacement candidate:

> **Generation = a discrete series of phase-lock levels at the same address** (ground generation = full lock; 2nd = partial; 3rd = marginal).

Under this candidate, mass ratios, linewidth ratios, and inverse-lifetime ratios between generations should correlate as deformations of a single distribution. Verification program (§19): the **generation triple-correlation test** — do mass ratio ≈ σ_ω ratio ≈ inverse lifetime ratio hold across generations? Confidence H (hypothesis; preliminary measurements support the mechanism).

11.4 The three mass readouts and (t, R, Q) — the component hypothesis (confidence H)

This series contains three readouts of mass: (i) $\mu\_\mathrm{Gram} = \det\Gamma/T^2$ (the degree of non-coherence [5]; T is the conserved readout R1); (ii) the relational quantity with the sea (divisor class gcd(m,16); Pillar 4; η = the charge axis); (iii) $m\_\mathrm{clock} = \langle\omega\rangle$ (the local clock mean; new pillar 8). Instead of viewing these as "different quantities under one name," we propose the **component hypothesis**: the three readouts correspond respectively to the three right-hand axes of the signature split of the zero closure, $x^2 + y^2 + z^2 = t^2 + R^2 + Q^2$ — **⟨ω⟩ = the t axis (proper clock), μ_Gram = the R axis (radial norm), sea relation = the Q axis (charge register)**. That is, the hypothesis is that **the internal state supporting the mass readouts has three components (t, R, Q)**. The observed mass (a Lorentz scalar) is the quadratic invariant $m^2 = M\_t^2 + M\_R^2 + M\_Q^2$ (equal to the spatial-side norm by the closure); there is no need to call the components themselves "three kinds of mass." The S row of Pillar 5 in which W/Z acquire mass by t winding (clock detuning) fits this picture in which different species draw their mass from different axes, and the sign-vector thought experiment [1] of the starting point returns at the readout level. A discriminating experiment is registered in §19.2: the **two-hypothesis form of the simultaneous three-way test** — (A) the dictionary hypothesis: the three lie on a single-valued curve; (B) the component hypothesis: the three are independent components lying on a quadratic-form surface. Discriminating point: if a pair of states can be constructed with nearly equal mass but different charge/clock character (the proton/neutron type — $\Sigma m = +3$ vs 0 with nearly equal masses), A breaks and B survives.

12. New pillar 9: the splitting readout — decay requires no added force

12.1 The claim

The splitting (decay) of a wave requires no force added to the dynamics. **Dephasing by non-uniform ω is itself the dynamics of splitting** — parts with different ω lose their mutual phase and become independent without anything being done. All that is needed is a readout criterion:

- **Particle number = the number of ω-locked equivalence classes**. Two regions are one particle while the accumulated relative clock phase stays below π (inside one interference fringe = coherent). π is a natural criterion, not a tuning parameter.
- Prediction: **$t\_\mathrm{split} = \pi/\Delta\omega$**.

This readout derives from the same fiber relational quantities as the spacetime gauges (τ_x, τ_t) and position. That is, **the classification of species, particle number, mass, lifetime, and splitting, and the gauges of space and time, are outputs of a single readout function** (the readout unification hypothesis). The perspective of classicality emerging by decoherence belongs to the lineage of [24]; this paper differs in formulating it as a readout criterion for particle number (the π criterion).

12.2 Measured (attachment #31; localization prototype)

Over six pairs of different composition, the readout detects the 1-particle → 2-particle transition, with **$t\_\mathrm{split}\cdot\Delta\omega/\pi = 1.29$ (CV 67%; all pairs at order 1)** (Fig. V2c).

Fig. V2c: The splitting readout. Accumulated relative phase crossing π, and $t\_\mathrm{split} = \pi/\Delta\omega$.

Known design limitation (honest registration): splitting was detected even for a same-composition control pair. The cause is that the phase of the sea carrier (wavelength n/3) differs between the packet positions, and the interference creates a spurious Δω. The refined experiment with carrier-phase-aligned placement (separation = integer multiples of the carrier wavelength) is registered in the verification program (§19).

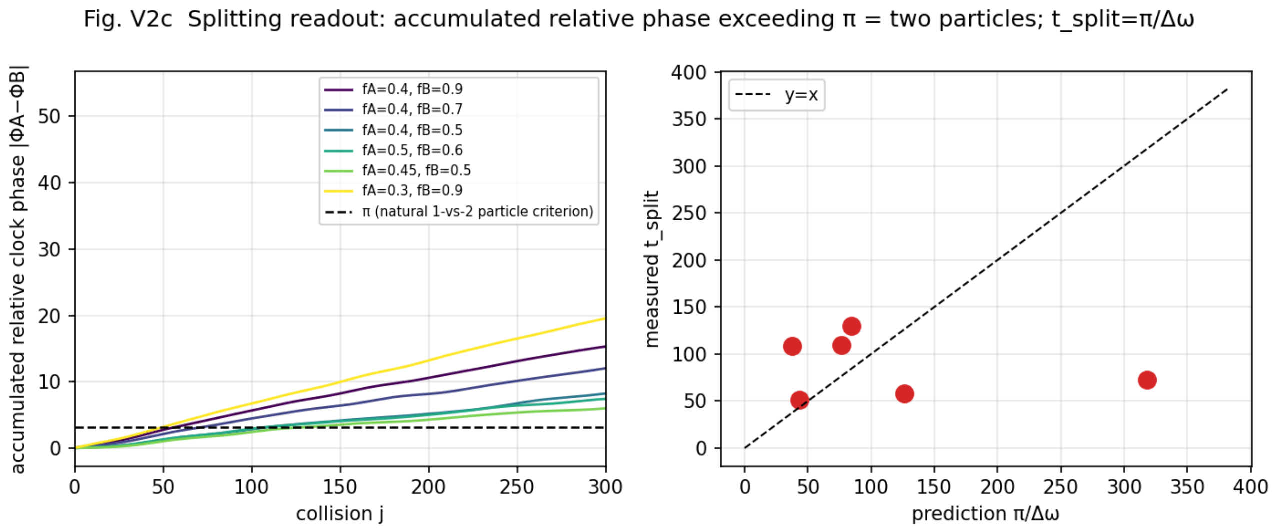

Figure 14: Fig. V2c

12.3 Side observation: survival of a coherent core

The self-overlap $C(t)$ does not fall to 0 but bottoms out at 0.55–0.76 — **a phase-locked coherent core survives**. This suggests that decay may be not "annihilation" but splitting into "a stable descendant + dephased fragments" (to be tracked).

13. New pillar 10: the two conditions of a stationary particle, and the hierarchy of existence

13.1 The two conditions (measured; attachment #32)

A true stationary particle of the dynamics requires both of:

1. **Localization**: stack all harmonics (even + odd) of the fundamental wavelength λ_0 with phases aligned at the center. Measured: the ringing (the maximum time-std of the instantaneous field τ_t) decreases monotonically with the harmonic width σ_k and approaches zero ($\sigma\_k=8$: 3.6×10^{-1} → $\sigma\_k=128$: 7.4×10^{-3}; Fig. V2d). With even harmonics only it plateaus at 4.0×10^{-1} — **both parities of harmonics are necessary**.
2. **Phase locking**: $\omega(x)$ = const (the proper clock uniform on the support). This is a nonlinear eigenvalue problem — the **candidate stationary-particle equation** of this model. The true next step is the enumeration of localized nonlinear eigensolutions $F[\Psi] = e^{i\Omega}\Psi$ of the universal interaction map F (§19.2) — if found, a "particle" becomes not a classification name but an isolated eigensolution of the dynamics.

Fig. V2d: The stationary-particle prescription. Full harmonics, center-phased → ringing → 0.

Accompanying exact structure: the channel ratio $\lambda = b/a = -i$ (circular polarization) is an **exact invariant manifold** of the elastic and inelastic rotations (verified algebraically; numerical deviation 0.0; amplitudes frozen everywhere) = the definite charge state. The construction by stacking the harmonic family (= child closures) originates in the two axioms of [15] (resolution + scale invariance), and the correspondence ringing = unstable relative equilibria connects to the onset-mode classification of [22].

13.2 The hierarchy of existence (the seating chart of the baryon hierarchy)

Combining new pillars 7, 8, and 9, the hierarchy of existence becomes one table (Fig. T, bottom panel; Table 3):

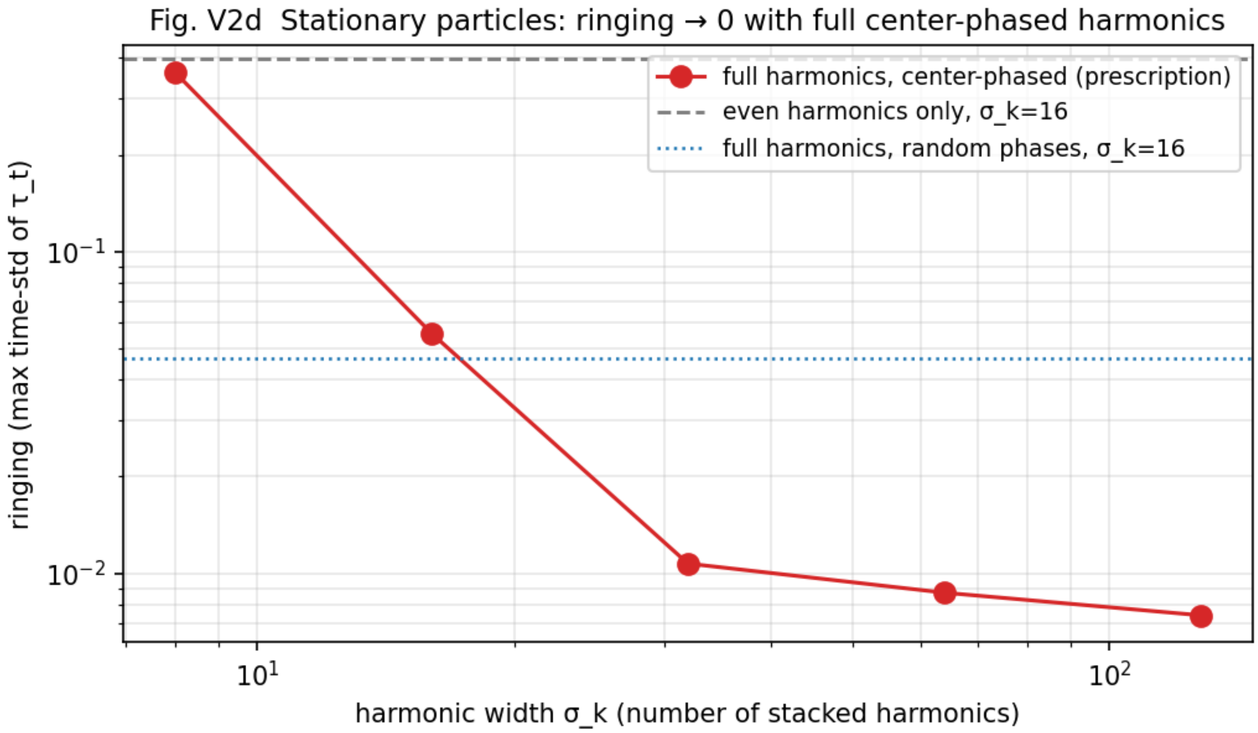

Figure 15: Fig. V2d

Level	Organizing principle	Charge	Statistics	Mass	Lifetime / decay	Conf.
sea (vacuum)	pure bosonic band (P_odd=0)	0	—	does not tick (exact fixed point, r=0)	∞	E
elementary species (62)	pure parity = antipodal two-point structure (no point localization)	Q=m/3	definite (cover 1/2)	⟨ω⟩	1/σ_ω	S
hadrons / baryons	ω-locked class with Σm≡0 (mod 3); p=uud:+3→+1 / n=udd:0→0	integer (automatic)	composite	⟨ω⟩ of the class	splitting = class bifurcation, t_split=π/Δω	S
classical bodies / condensates	equal parity mixture (all harmonics, point-localized)	integer	none (mixed)	Σ⟨ω⟩ (gravity source)	macroscopic	S

Baryons are not “an extra box outside the 62 species” — they hold seats for mass, lifetime, and decay as one level of the same readout, the ω-locked classes. The integer-charge check of hadrons of Pillar 2 (p = uud → +1) lives directly in the third row of this table.

14. The consistency battery — the classification is not an artifact of a particular engine

In the v2 verification program we constructed two localization prototypes of the θ readout of the two-body system [3] (§2.4). Does localization break the classification? — nine pillar experiments (v4b, v6, v7, v9, v10b, v13b, v14, v15, v17b) were run **with script bodies unmodified**, swapping only collision_step, against three engines in parallel (the published system = global θ / local θ sharp / local θ smooth; the attached consistency-battery folder).

Results:

1. **Soundness**: the global-θ runs agree completely with the published result JSONs for all scripts (verdict strings and all numbers).
2. **The integer and phase structures are completely robust**: the Z_2 covering cross table (identical verdicts in 4 cells), the neutrino row (covering 2; Qz2 = 1.00; phases strictly $\{0, \pm\pi\}$), confinement (isolated exactly 0, $\sim 10^{-29}$), the ledger identity ($\Delta Q3 = -3\Delta W$ residual $\sim 10^{-14}$), address equivalence (all claims identical), the divisor-class theorem (pass; spreads even shrink). **The skeleton of the classification is invariant under localization of the readout.**
3. **Continuous quantities shift by a few percent**: the equilibrium fermionic fraction of the sea f* is 0.4695 (published) → 0.4578 (sharp, −2.4%) / 0.4428 (smooth, −5.6%); hadronization rates by −2.7 to −4.1%. Continuous quantities require recalibration if a local engine is adopted.
4. **The single qualitative regression = Q_wind conservation**: the conservation of the net winding charge, exact under global θ (CV 2.6×10^{-7}), leaks slowly under local θ (5–6% over 4000 collisions). This is a limitation of the prototypes and is registered as requirement 10 for the universal function (§19).

This battery is simultaneously independent evidence that the classification of Part I is carried not by “the particular implementation called global θ” but by the structure of the readout (band fractions, parity, mod 3).

Part III — Position, weaknesses, refutations, and requirements for the next paper

15. Relation to prior work, and the novelty restricted

Each component of this hypothesis has a strong prior lineage: winding = charge (Skyrme [17]; Kaluza–Klein [18]), mod-3 confinement (triality [10][11]), the double cover of statistics (Finkelstein–Rubinstein [12]), vacuum = sea (Wen, Volovik [13]), gravity = square (KLT/BCJ [16]), full-assignment enterprises (the Eightfold Way [9]; preons [14]), the linewidth-lifetime relation (Weisskopf–Wigner [23]), classicality by decoherence (Zurek [24]). **This paper claims no novelty for the components.** The novelty is restricted to four points:

1. That these components come out simultaneously as **measurements of a single anonymous dynamics** that assumes no particles (confluence).
2. The presentation of a counterpart in which charge unit, confinement, and elementary-charge uniqueness come from **the divisor structure of the observation clock (the readout side)**, not from axioms on the dynamics side (the center of a gauge group).
3. The measurement that the double cover of statistics lives in **the clock** (state period = observation period × 2), not in space (rotations of configuration space) (with v2 adding its

spatial embodiment as antipodal structure; §10).
4. (v2) The presentation of the unification in which mass, lifetime, generations, and splitting come out as statistics (mean, width, locked classes) of a single clock field ω(x), from the same readout function as the spacetime gauges.

16. Weaknesses and untested parts (honest registration)

1. **The origin of generations is unresolved (though v2 obtained a positive candidate)**: the naive hypothesis generation = χ-band position is refuted (χ-band translation symmetry; §6.1). The v2 candidate is the discrete series of phase-lock levels (§11.3) — verification program specified; not yet run.
2. **The W/Z rows**: not directly producible in the low-energy laboratory ($N \le 16$, small sea; immediate register-edge decay; structural). The $1/\delta^2$ scaling test of the effective contact vertex (the Fermi-theory road) is specified; not yet run.
3. **The graviton row**: the 2ω quadrupole line exists at all N, but the pre-fixed criterion ($|\text{ratio}-2|<0.02$) passes only 1/3 — frame calibration incomplete.
4. **Dependence on the register order ne=16**: details of Table A (neutralization steps etc.) are functions of ne. Re-examination on ne-variable systems is needed (also a testable prediction: "the period of the periodic table is a function of the register order").
5. **The unification of Q_em = m/3 with the transmission charge √(4πα) [6][19] is incomplete** — the relation between the "unit" (3 windings) and the "value" (0.302822) of the elementary charge is a task for the next papers. v2 addendum: the hypothesis "fermions concentrate at odd fraction $f \approx 0.7 = 1-\sqrt{4\pi\alpha}$" was **refuted** by test (the f of actual species distributes over 0.02–0.53; the prototype's maximum position is configuration-dependent; §18 (14)). The seat of 0.6972 remains in the context of **the reflection rate of a charge-readout event**, not species composition.
6. Part of the sea constructions (artificial projected seas) are known to be outside the conservation class (Nyquist fold-back; §4.2). The quantitative results of the affected experiments are treated with reservation; the qualitative conclusions were cross-checked on in-class series.
7. **(v2) The quantitative values of Part II are prototype measurements**: $\tau \cdot \sigma_\omega = 32.1$, the t_split coefficient 1.29 etc. carry scatter of CV 30–67%. The claims are restricted to scaling forms (invariance of products; order-1), and no precise coefficient values are claimed. The splitting control requires carrier-phase alignment (§12.2; a design rule).
8. **(v2) The relation of the two clocks is unresolved**: the local reflection clock of the two-body system (τ_t; stops at $P_{odd}=0$) and the collective phase clock of the N-body condensate ($\omega_{clock} = \pi/72$; Pillar 1) are not yet unified (§9.2). If unified, Pillar 1 and new pillar 6 connect; if not, the word "clock" was naming two independent quantities — registered as a decidable open problem.

17. Refutation conditions

1. If an observation clock with a denominator other than 3 is constructed that rectifies the entire product set of the same charged walk into a unique elementary-charge unit, Pillar 2 is rejected.
2. If, in the long-window limit with the same condensed phase, the same update rule, and the same settling criteria, a series is obtained in which ω deviates systematically from π/72, Pillar 1 is rejected.
3. If a species is constructed whose covering degree varies independently of χ parity, Pillar 3 is rejected.
4. If a measurement shows species properties (mass, lifetime, polarization) differentiating without a sea, Pillar 4 is rejected.
5. If a construction is found in which a species with $m \neq 0 \pmod 3$ becomes readable alone, the confinement law is rejected.
6. (v2) If a parity-symmetric dynamics (r symmetric in parity) reproduces any of the covering-degree difference, confinement, or elementary-charge uniqueness, new pillar 6 is rejected.

7. (v2) If a point localization of a pure-parity species is constructed, new pillar 7 is rejected.
8. (v2) If a family is found in which $\tau_{coh}\cdot\sigma_\omega$ systematically diverges or vanishes, new pillar 8 is rejected.
9. (v2) If, after carrier-phase alignment, $t_{split}\cdot\Delta\omega/\pi$ does not converge to order 1, the π criterion of new pillar 9 is rejected (search for another criterion).

18. Record of refutations and corrections (15 items, summarized)

We record every hypothesis and instrument refuted in the course of this work: (1) short-window sideband addresses are transients, not stable species (vanish in the long window). (2) The mass = detuning^2 law is an apparent relation on transients. (3) The naive form of ±1 fixed-point uniqueness (any pure m is a fixed point; uniqueness does not follow). (4) The mass-width correlation fails ($r=-0.07$). (5) The sea-construction-convention hypothesis (χ analytic projection) fails — the true cause of conservation breaking is Nyquist fold-back (§4.2). (6) ±1 uniqueness by escape rates is undecidable in principle by the divisor-class theorem. (7) The kinematic version of the SU(2) recurrence-angle discriminator is spinorial for all states (trivial). (8) Single-point covering-degree judgment picks accidental π proximity (calibrated to recurrence-sequence judgment). (9) The linear $\ell=2$ frame for the graviton is absent (reinterpreted as composite). (10) Generation = χ-band position is refuted. (11) The band-shift method of W/Z detuning injection fails (effectively linear dispersion). (12) The projection method of the ν construction fails (the bundle hair m=+1 is the cause; solved by the shift method). (13) The earlier projected seas are amplified numerical residue (legitimate as an m=0 field; qualitative results unchanged; recorded). **(14) (v2) The hypothesis "fermions concentrate at odd fraction f≈0.7" is refuted** — the prototype's signal maximum at f≈0.77 is configuration-dependent (vanishing under amplitude/harmonic-width change), and the f of actual species distributes over 0.02–0.53, not clustering at 0.7; dynamical equilibria are pulled instead toward f≈0.5 (near symmetry). **(15) (v2) The window-mean drift indicator of the splitting experiments is broken** — it misjudges stationarity through incommensurability of beat periods with the observation window (calibrated to the true indicator = the time-std of the instantaneous field; §13.1). Details of each item are in the attached analysis notes.

19. Requirements handed to the unified dynamics (the specification of the next paper; 8 → 11 items)

To verify the classification hypothesis at the dynamics level, the two dynamics must be unified into a single **universal interaction function**, and the 62 species must be shown to be generated, to decay, and to interact as its spectrum. The present measurements fix the requirements that function must satisfy:

1. **Anonymity**: only wave shapes as input; no per-species casework (no IF statements).
2. **All channels at once**: electromagnetic (winding exchange), weak (clock detuning), strong (mod-3-unreadable raw-winding residue), and gravitational (second-moment coupling) arise not as separate terms but as different readouts of one operation. It must drive the three planes (xy = spin / zR = mass / tQ = charge) [20] simultaneously.
3. **Conservation laws automatic**: cyclic winding conservation (§4.2), closure conservation, and norm conservation held by the structure of the operation, not by design.
4. **Reproduction of the universal clock**: the U^{144} clock lattice with $\omega = \pi/72$ (§3) appears as spectrum.
5. **Emergence of selection rules**: the sum rule $m^*=2m_B-m_s$, the doubling/inversion ladder, Z_2 phase quantization, and the divisor-class structure appear without hand placement.
6. **Exact computability**: anonymity ⇒ computation linear in state size [4] (closed-form or integer-exact).
7. **Specialization to the two systems**: the two-body closed-form solution [3] and the N-body relational-wave dynamics [2] re-derived as restrictions/limits of the function.
8. **Refutation form**: if any of the 62 assignments (§7.2) fails to appear in the spectrum, the periodic-table hypothesis is rejected row by row.

9. **(v2) Gravity on the gauge side**: gravity cannot be inserted as a state-side potential — it necessarily breaks the closure $\Sigma x^2=0$ (measured: an injection of amplitude 10^{-3} breaks it by 31% in 200 steps; the bare dynamics 10^{-13}; gauge-side operations exactly 0). Gravity enters as gauge distortion on the projection (readout) side that creates space.
10. **(v2) Conservation under localization**: the localization of the θ readout must preserve Q_wind conservation exactly (§14: both current prototypes leak 5–6% over 4000 collisions = requirement unmet; conservation-compatible localizations such as the circular-polarization invariant manifold are the search direction).
11. **(v2) Splitting as readout**: splitting/decay is not added as a force but read as ω-coherence clustering of the spacetime readout function (the π criterion; no tuning parameters; the interaction function untouched; §12).

19.2 The verification program (8 items, in priority order)

In parallel with the construction of the universal function, we register the experiments that can directly strengthen or refute the present hypotheses:

1. **Dynamical derivation of denominator 3, and charge response**: the dominance of denominator 3 currently rests on the locking measurements [20]. An analytic derivation of why the denominator-3 Arnold tongue is maximal in this dynamics would promote the "3" of Q_em = m/3 from empirical choice to dynamical necessity. Together, measure whether the exchange and phase shift of two species (m_1, m_2) scales as the product $m_1 m_2/9$ (the Coulomb-type charge-product test) — if it passes, m/3 is promoted from bookkeeping to a physical charge appearing in interactions.
2. **Clock covering degree and exchange sign in one experiment (the decisive experiment to run before the universal function is built)**: connect covering degree 2 ⟺ exchange sign −1 of two identical species in a single experiment. If the exchange operation E and the one-clock-cycle operation T are shown to belong to the same nontrivial class of one double cover on state space ($E \simeq T$, $E^2=T^2=I$), then half-integer return, exchange sign, exclusivity, and clock covering reduce to a single Z_2, and Pillar 3 advances from "resembles statistics" to "a reconstruction of the structure of statistics."
3. **(v2) The simultaneous three-way mass-dictionary test (two-hypothesis form; §11.4)**: on one family of states, measure μ_Gram = $\det\Gamma/T^2$, r = P_odd/P_total, and ⟨ω⟩ simultaneously, and discriminate whether they (A) lie on a single-valued dictionary curve or (B) lie on a quadratic-form surface (the component hypothesis). Under A, the mass theory of the previous paper, new pillar 6, and new pillar 8 contract into a single theorem candidate; under B, the (t,R,Q) three-component structure of the mass readouts is established — a discriminating experiment that fixes structure whichever way it falls, and with this one experiment the Gram mass of the previous paper, the sea-dependent mass of v1, the clock mass of v2, and the (t,R,Q) hypothesis are sorted at once; hence its priority is raised.
4. **The deformation law of the table under variable register order**: for ne = 12, 18, 24, 32, …, measure whether divisor classes, neutralization steps, readability, and covering structure track gcd(m, ne)/Div(ne). If they track, the divisor-class theorem generalizes; if not, there is hidden structure specific to 16.
5. **Blind prediction**: without Standard-Model names, predict from the model alone the unoccupied addresses, lifetime ordering, charges, covering degrees, and decay targets, then compare after the fact with the known particle table — a test format with the decisive power of the Ω^- of the Eightfold Way.
6. **(v2) The generation triple-correlation test with orthogonal control**: construct the phase-lock-level series at one address and measure whether mass ratio ≈ σ_ω ratio ≈ inverse-lifetime ratio across generations (the direct test of new pillar 8). Together, with a factorial sweep of amplitude, composition, and harmonic width, construct series in which ⟨ω⟩ is held nearly constant while σ_ω varies (and vice versa), to confirm that the +0.904 mass-linewidth correlation is not a product of common driving.
7. **(v2) The carrier-phase-aligned splitting experiment**: align packet separations to integer multiples of the carrier wavelength, separate the composition-borne Δω from the spurious

carrier-phase Δω, and re-test t_split = π/Δω (the refinement of new pillar 9).

8. **(v2) Explicit verification of the Z_2 covering on the local engines**: re-run the covering-degree experiments not yet included in the consistency battery (§14) on the local engines, completing the robustness of the integer structures.

20. Reproducibility

All experiments are published as deterministic scripts (32) in the same folder as this paper. The correspondence table (the experiment-inventory file, including the v2 additions #29–35) gives the complete mapping of scripts ↔ result JSONs (29) ↔ figures (the ten v1 figures in Japanese and English; the four new v2 figures in Japanese and English; the periodic-table visual in Japanese and English) ↔ claims. The full consistency battery (§14; 27 runs = 9 experiments × 3 engines) is published in a separate folder (the attached consistency-battery folder) together with run logs, result JSONs, the comparator, and the battery report. Judgment criteria are fixed and recorded at the head of each script before execution.

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Note: S. Coleman, "Quantum sine-Gordon equation as the massive Thirring model," Phys. Rev. D 11, 2088 (1975) is referenced via [7] as the representative of the bosonization lineage.